

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

May 2012

EC312 Digital Signal Processing

Date: 11.05.2012, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define the term "digital signal processing" with its basic block diagram. Differentiate the multi channel signal and the multi dimension signal. [05]
- (b) Determine causality and time variance properties of Up Sampling (expansion) system. [02]
- Q - 2 (a) Describe the stability and causality criteria for Linear Time Invariant System. [05]
- (b) Determine IZT for $H[z] = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$ [05]
- (i) ROC $|z| > 1$, (ii) ROC $|z| < 0.5$
- (c) Prove any two Z-transform properties with example as given below: [04]
- (i) Time reversal, (ii) Linear, (iii) initial value theorem

OR

- Q - 2 (a) Explain the different windowing techniques for FIR filter designing. [05]
- (b) (i) Define the transposition theorem with example. [05]
- (ii) Find 4 point IDFT of $X(k) = \{3, 1\}$
- (c) List out the properties of DFT and prove any one. [04]
- Q - 3 (a) Find the convolution using mathematical method for system which having impulse response $h[n] = \{1, 2, 1, -1\}$ and input $x[n] = \{1, 2, 3, 1\}$ [04]
- (b) Explain the digitization process with necessary figures. Find the Nyquist sampling rate for signal $x(t) = 2\sin 480\pi t + 3\sin 720\pi t$ [06]
- (c) Explain all pass system with its typical pole and zero. [04]

OR

- Q - 3 (a) Realize following system transfer function using direct form I and direct form II. Also draw the signal flow graph for the same. [07]

$$H[z] = \frac{1 + 0.2z^{-1} - 0.2z^{-2}}{1 - 0.2z^{-1} + 0.3z^{-2} + z^{-3}}$$

(b) Prove (i) symmetry property

[05]

$$W_N^{k+\frac{N}{2}} = -W_N^k$$

(ii) Periodicity property

$$W_N^k = W_N^{k+N}$$

(c) Give the importance of "Zero Padding" in the discrete Fourier transform computation.

[02]

SECTION - II

Q - 4 (a) Explain Signal ambiguity in frequency domain (aliasing). How this aliasing effect is removed?

[03]

(b) Compute DFT of signal $x[n] = \{0, 1, 2, 3\}$

[04]

Q - 5 (a) Compare the characteristics of IIR and FIR filter and Draw the parallel realization of

[07]

$$\text{filter } H[z] = \frac{1 - z^{-1}}{1 - 0.2z^{-1} - 0.15z^{-2}}$$

(b) Draw and explain radix 2 decimation in time FFT implementation signal flow graph (butterfly diagram) for 8 point FFT.

[07]

OR

Q - 5 (a) The direct form II realization for IIR filter is given in below figure:

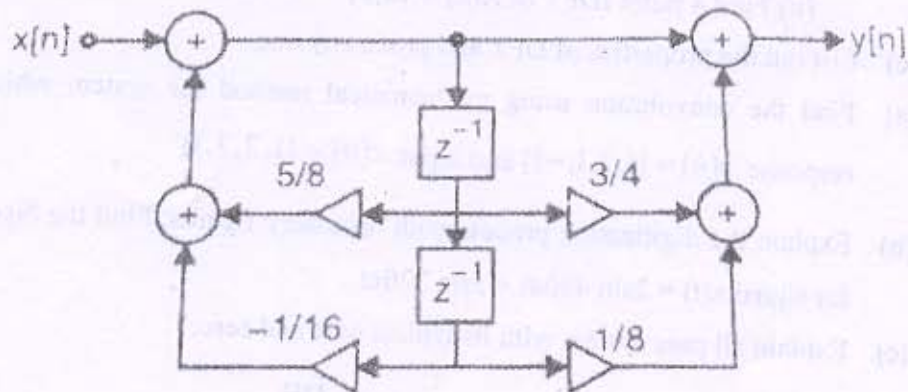
[07]

(i) Obtain transfer function $H[z]$

(ii) Obtain difference equation.

(iii) Realize it using cascade form with first order sections.

(iv) Draw pole zero plot for given filter.



(b) (i) determine $r_{xy}[l]$ and $r_{yx}[l]$ for given sequence

[07]

$$x[n] = \{-3, -2, 1, 4, 8, -3\} \quad y[n] = \{1, 1, 1, -1, 2, -2\}$$

(ii) An analog signal $x(t) = 4 \cos 100\pi t$. Calculate:

(a) Minimum sampling rate to avoid aliasing.

(b) If the signal is sampled at the rate $F_s = 200\text{Hz}$, what is the discrete time signal after sampling.

Q - 6 (a) Compute (i) linear convolution, (ii) circular convolution of give sequences using graphical method $x_1(n) = \{1, 2, 3, 2\}$, $x_2(n) = \{3, 2, 3, 1\}$ [07]

(b) Explain the impulse invariant method for IIR filter designing with example. [07]

OR

Q - 6 (a) List the applications of DSP with examples. Explain following terms: [07]

(i) deterministic and random signal, (ii) energy and power signal (iii) continuous value and discrete value signal

(b) Determine the impulse response and the step response of the following causal systems. [07]
Plot the pole-zero patterns and determine which of the systems are stable

(i) $y(n) = \frac{3}{4}y(n-1) - \frac{1}{8}y(n-2) + x(n)$

(ii) $y(n) = y(n-1) - 0.5y(n-2) + x(n) + x(n-1)$

(iii) $H[z] = \frac{z^{-1}(1+z^{-1})}{(1-z^{-1})^3}$
